

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Due Date	30/3/26		

Experiment 9: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

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1 Objective

To implement and compare K-Means, DBSCAN and Hierarchical clustering on HAR dataset.

2 Dataset

- Activities: Walking, Upstairs, Downstairs, Sitting, Standing, Laying
- Sensors: Accelerometer + Gyroscope
- Samples collected from 30 subjects

3 EDA

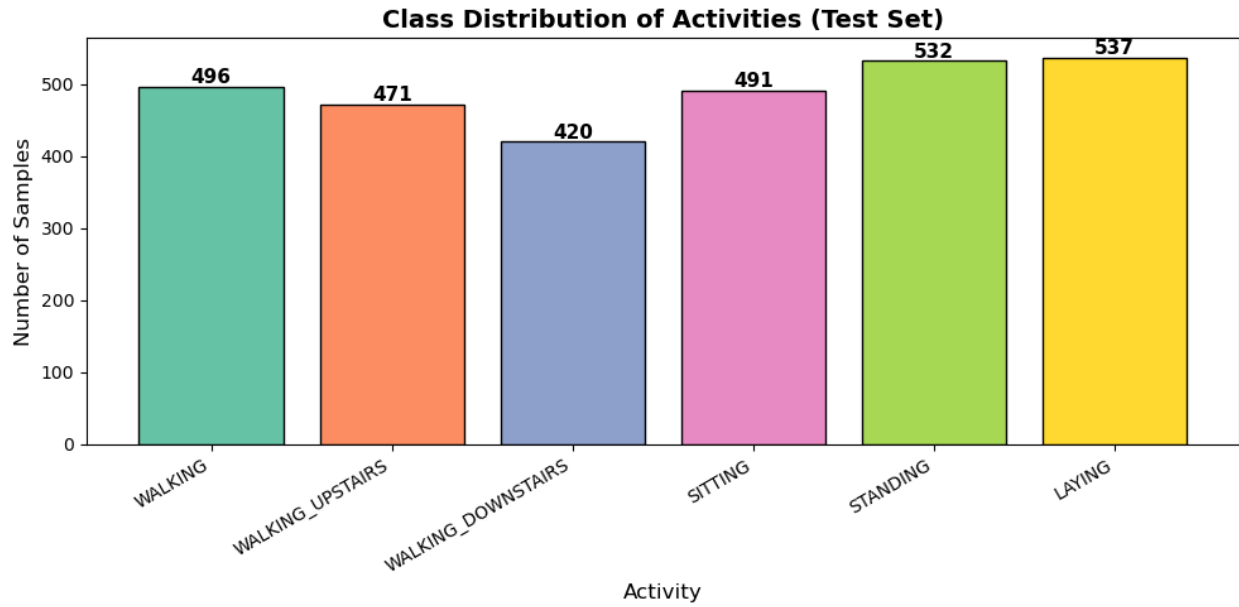


Figure 1: Class Distribution

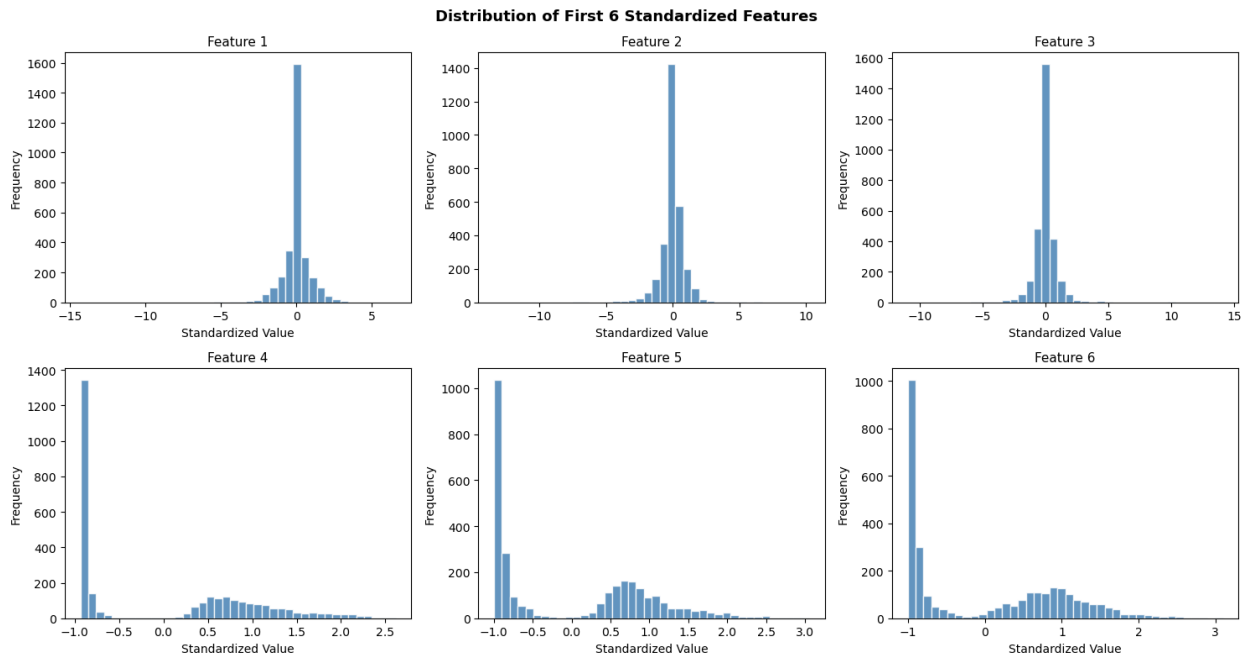


Figure 2: Feature Distribution

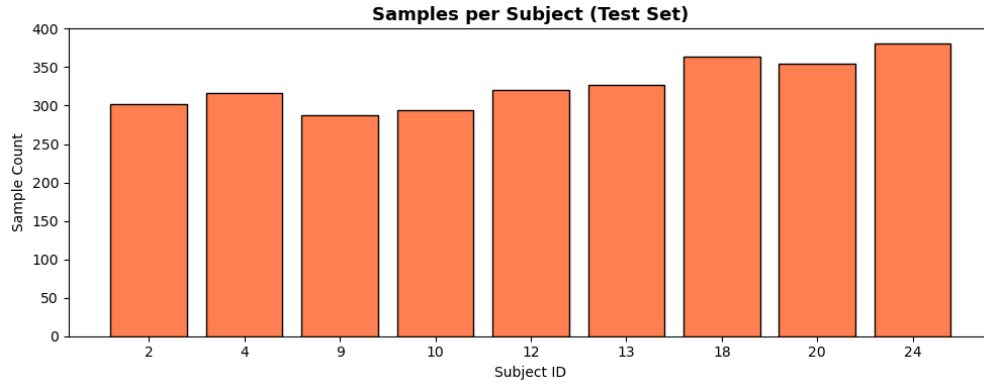


Figure 3: Samples per Subject

4 Dimensionality Reduction

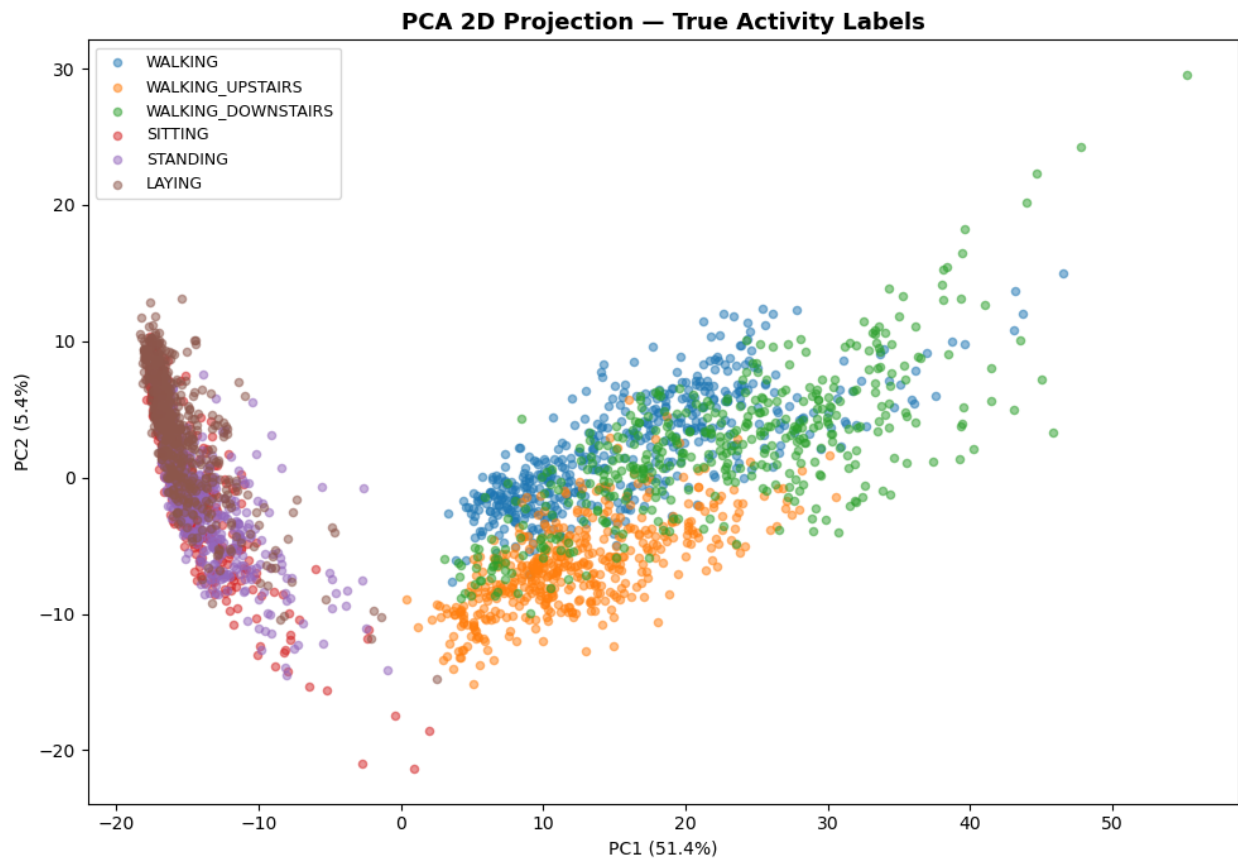


Figure 4: PCA Projection

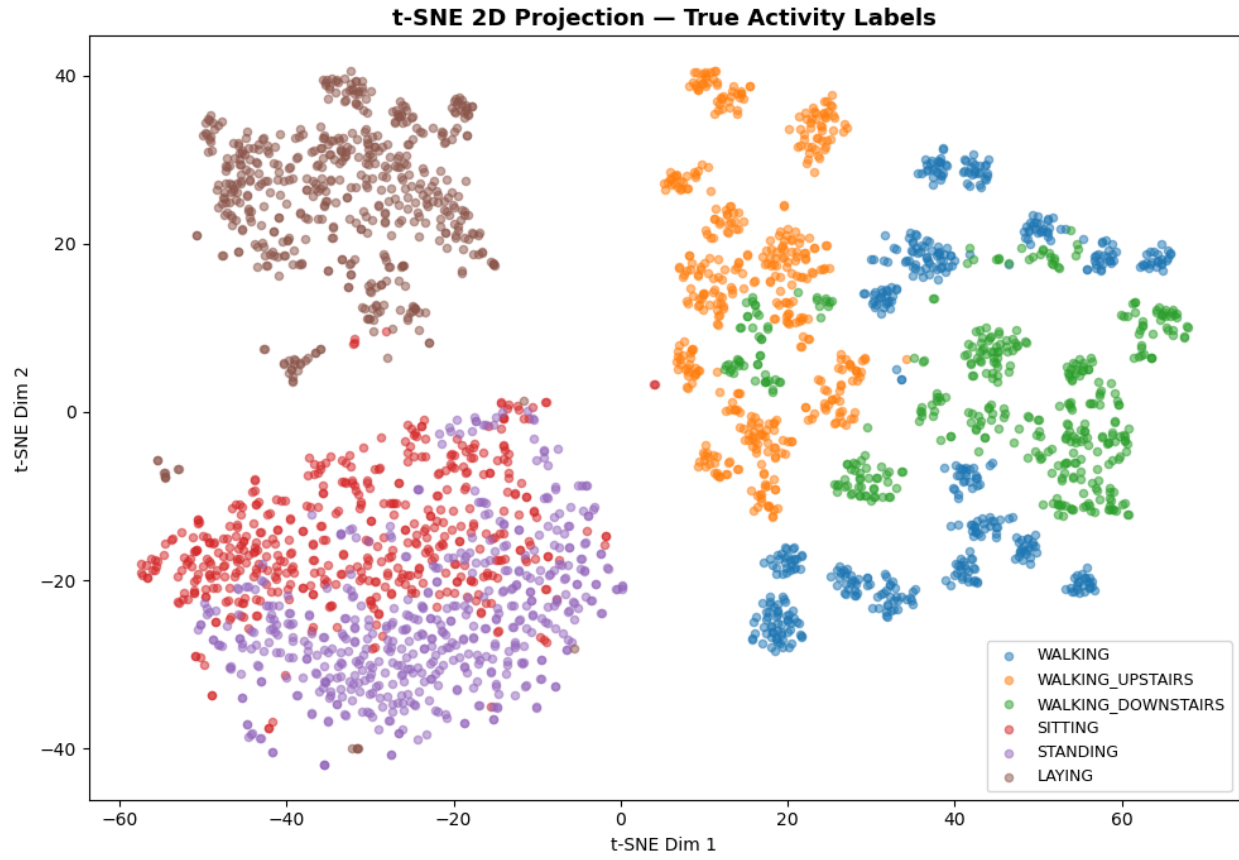


Figure 5: t-SNE Projection

5 K-Means Clustering

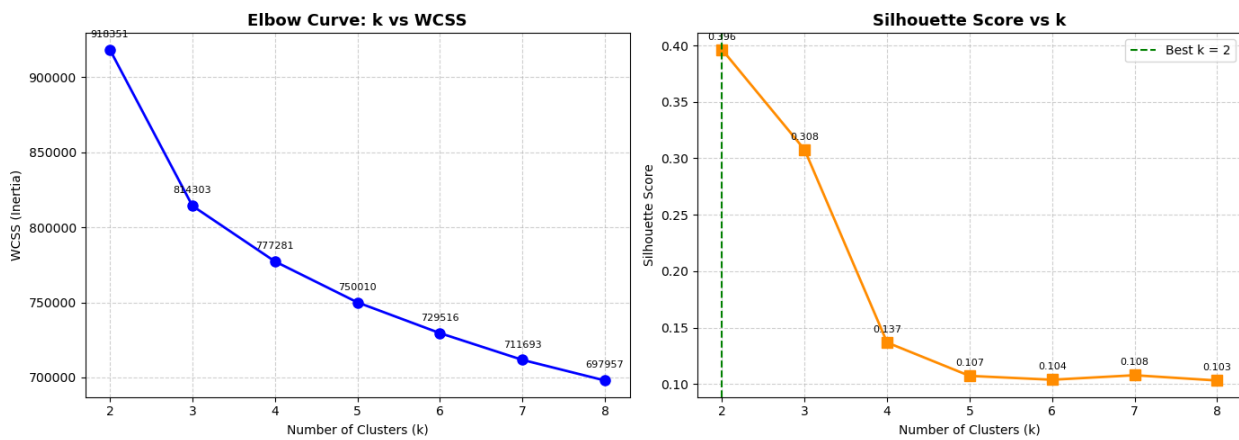


Figure 6: Elbow Curve and Silhouette Score

Table 1: K-Means Elbow Method Results

k	WCSS	Silhouette Score
2	918351.30	0.3963
3	814303.31	0.3076
4	777281.30	0.1368
5	750010.25	0.1072
6	729516.26	0.1038
7	711693.28	0.1078
8	697957.15	0.1031

Best $k = 2$ (highest silhouette score)

6 K-Means Results

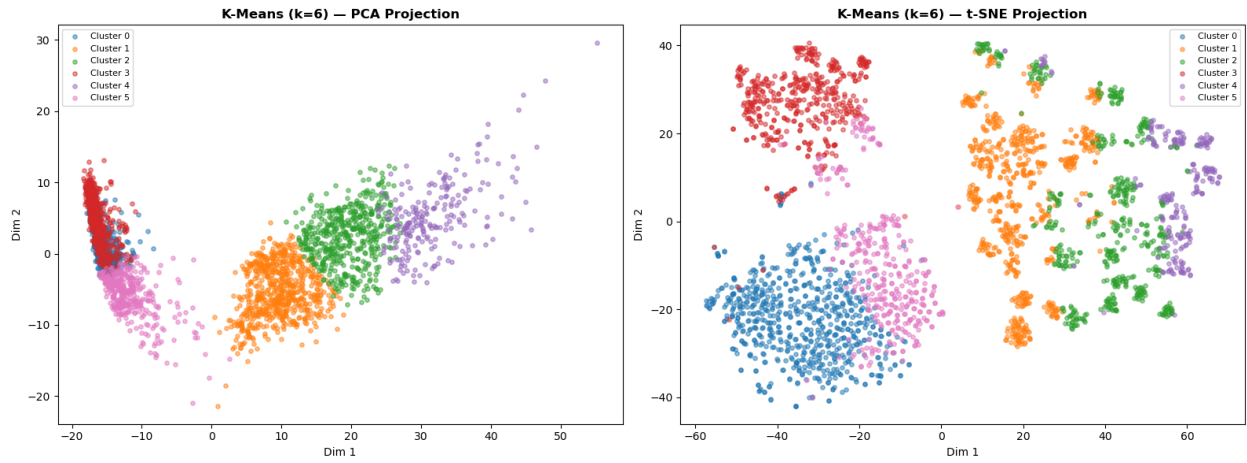


Figure 7: K-Means Clusters (PCA & t-SNE)

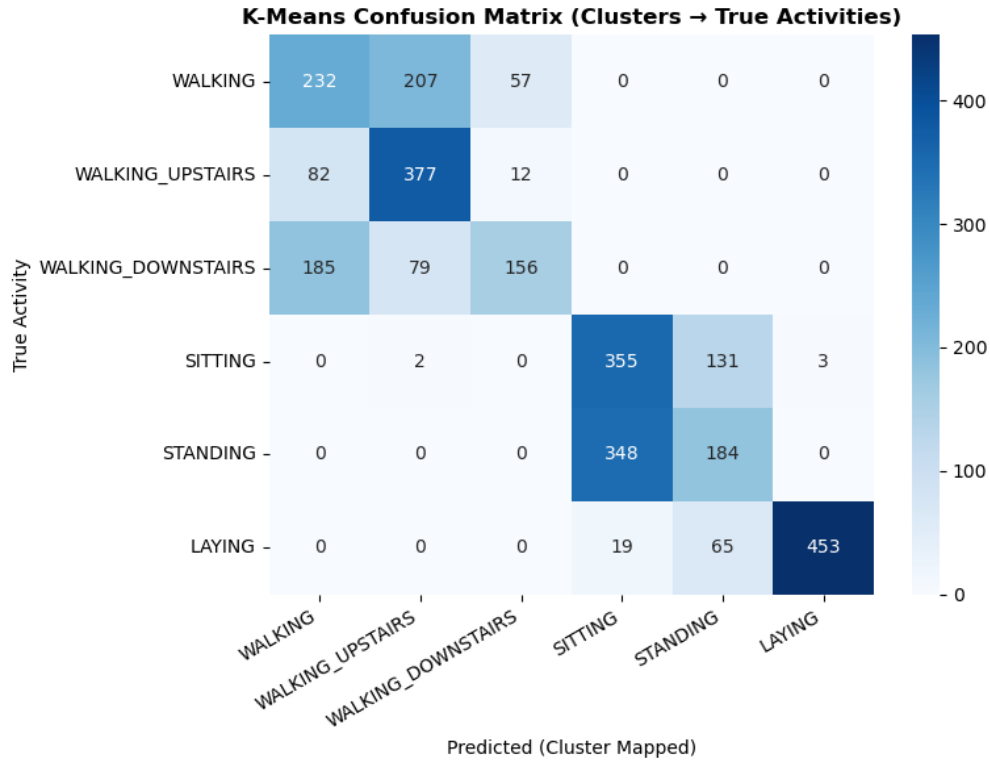


Figure 8: K-Means Confusion Matrix

7 DBSCAN

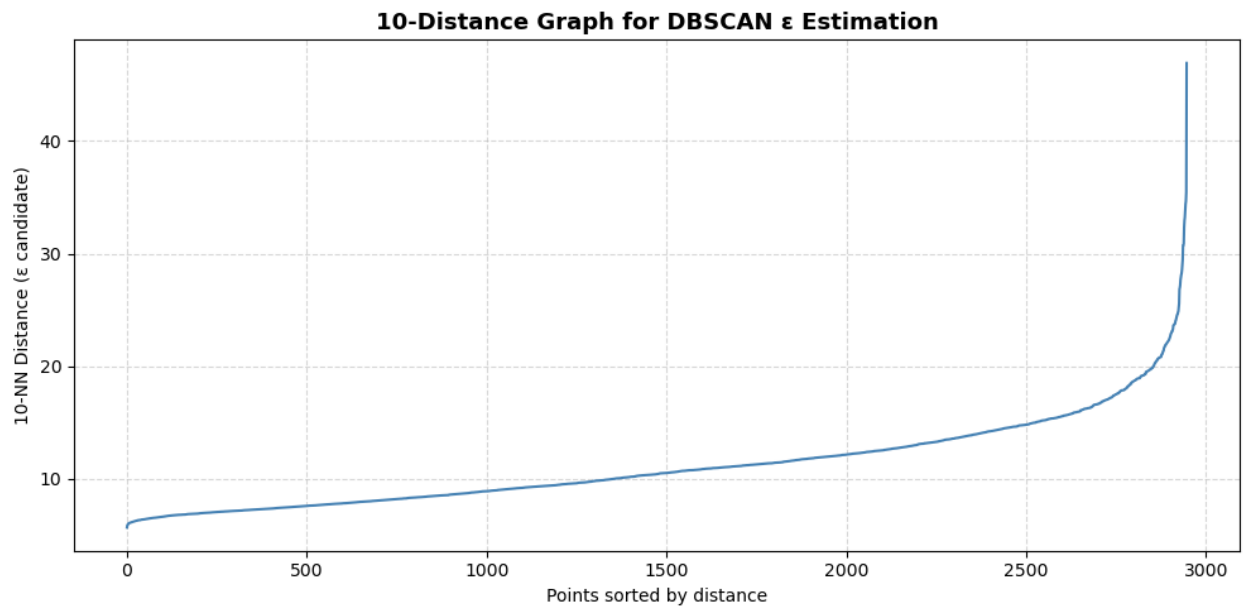


Figure 9: Epsilon Estimation

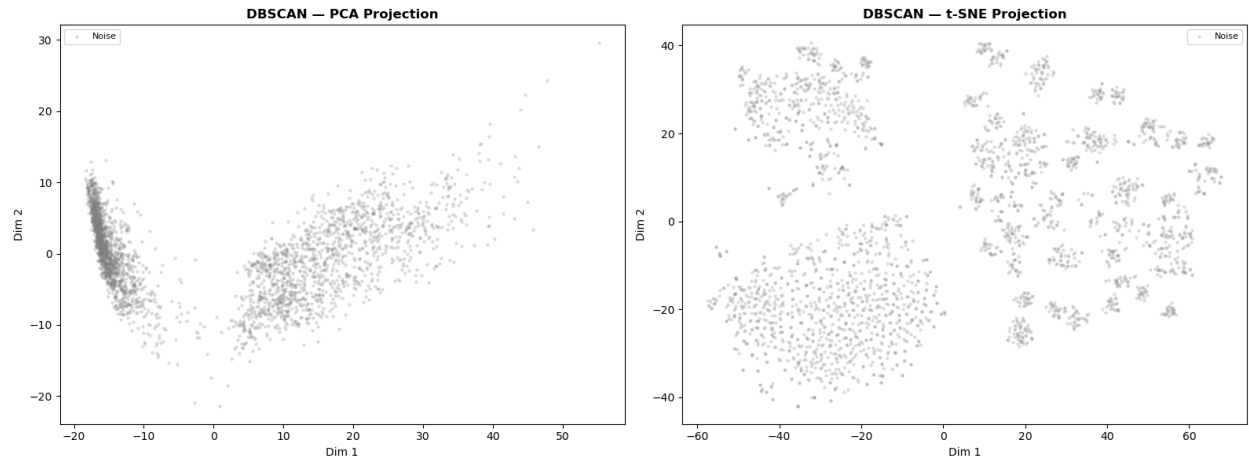


Figure 10: DBSCAN Clusters

8 Hierarchical Clustering

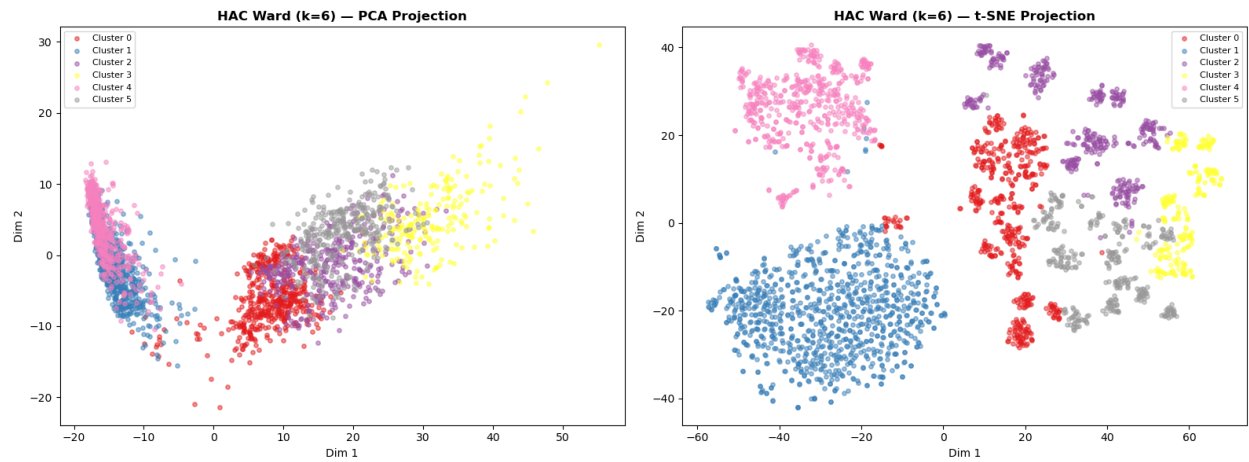


Figure 11: Dendrogram

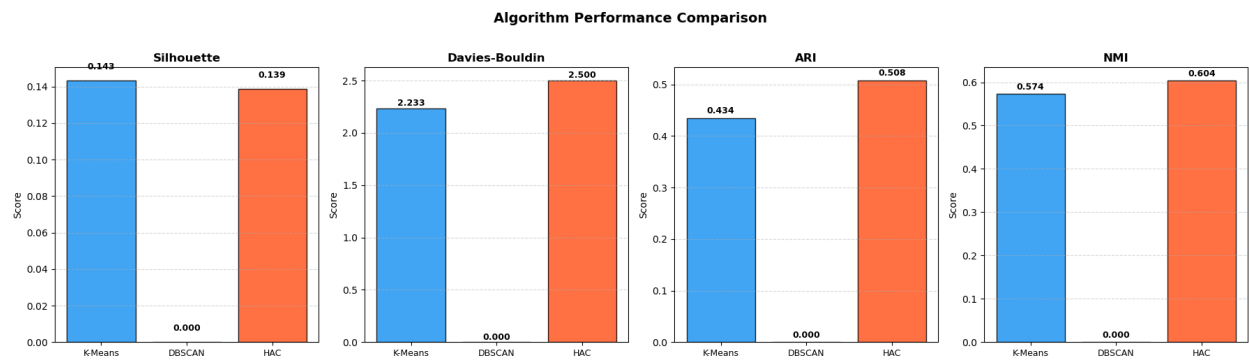


Figure 12: HAC Clusters

9 Performance Comparison

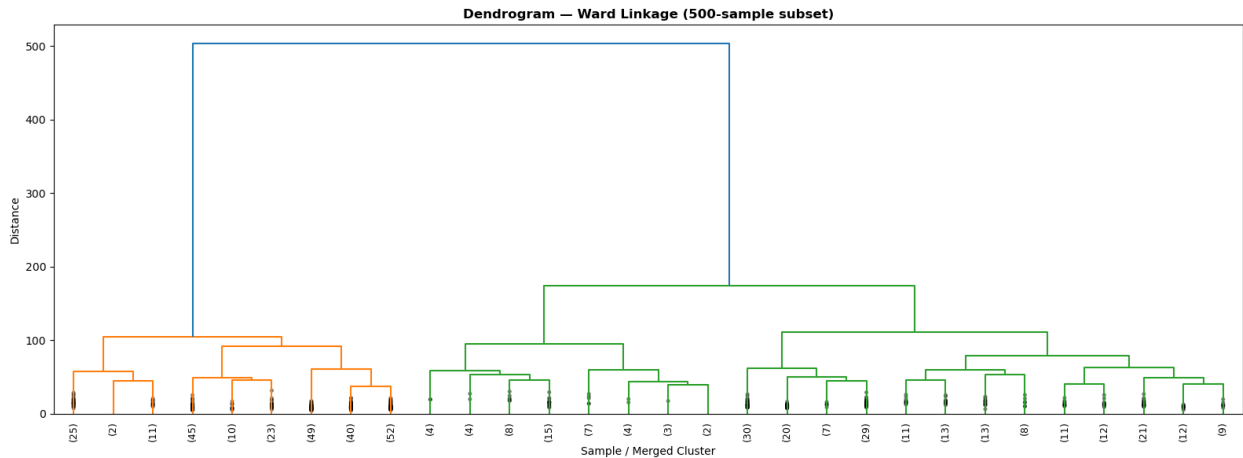


Figure 13: Algorithm Comparison

Table 2: Clustering Metrics

Model	Silhouette	DB Index	ARI	NMI
K-Means	0.143	2.233	0.434	0.574
DBSCAN	0.000	0.000	0.000	0.000
HAC	0.139	2.500	0.508	0.604

10 Observations

- K-Means performed well but depends on k.
- DBSCAN failed due to parameter sensitivity.
- HAC gave best clustering (highest ARI, NMI).
- PCA and t-SNE improved visualization.

11 Conclusion

Clustering algorithms were compared on HAR dataset. HAC performed best overall. K-Means was effective but required tuning. DBSCAN struggled with parameter selection.